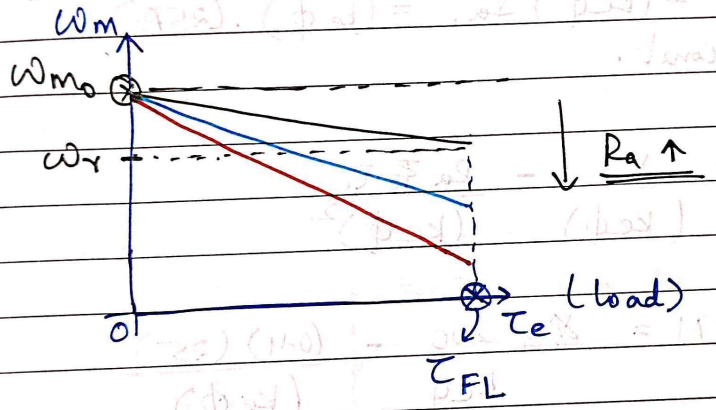
$$\omega_m = \underbrace{\frac{V}{k_e\phi}}_{\omega_{m_0}} - \underbrace{\frac{R_a T_e}{(k_e\phi)^2}}_{\text{Load dependent.}} \qquad T_e = k_e\phi . \underbrace{I_a}_{\text{Load}}$$

Speed Torque Characteristics. (ω_m v/s T_e)



ω_r : rated speed at full load.

SEDC Motor : $\Delta\omega = \omega_{m_0} - \omega_r \simeq 0$.
Constant speed motor.

Full load / Rated Speed.

Q 200V, 500RPM DC motor has $R_a = 0.4\Omega$. $I_{FL} = 2$
If a resistance $R = 1.6\Omega$ is added to armature ckt, find speed of motor while delivering the same load torque?

$T_e = (k_e\phi) I_a = (k_e\phi) . (25A)$
$\rightarrow$ const.

$$\omega = \frac{V}{(k_e\phi)} - \frac{R_a T_e}{(k_e\phi)^2}$$

$$500\,RPM = \frac{200}{k_e\phi} - \frac{(0.4)(25)}{(k_e\phi)}$$

$$= \frac{190}{(k_e\phi)}$$

$$\omega = \frac{200}{(k_e\phi)} - \frac{(T_e)(0.4+1.6)}{(k_e\phi)^2}$$

$$\omega = \frac{200}{(k_e\phi)} - \frac{(25)(2)}{(k_e\phi)} = \frac{150}{k_e\phi}$$

$$\frac{\omega}{500} = \frac{150}{190} \qquad \omega = \frac{(500)(150)}{(190)}\,RPM$$

Why load = ½ ? Power ½ or ½ when halved?

$\omega = 394.74$ RPM.

$E_{b_1} = V - R_a I_a = 200 - (0.4 \times 25) = 190\,V$

$E_{b_2} = V - (R_a + R_s) I_a = 200 - (2 \times 25) = 150\,V$

$\frac{N_2}{N_1} = \frac{\omega_{M2}}{\omega_{M_1}} = \frac{150}{190} \Rightarrow N_2 = \frac{150}{190} \times 500.$

$\simeq 394.73$ RPM

(1) What will be the speed at half the full load

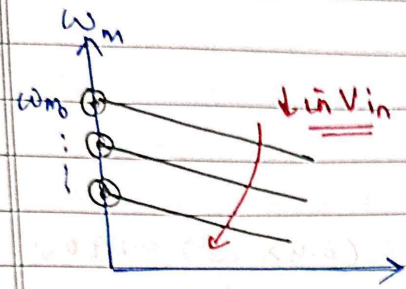
(2) For $R_s = 1.6\,\Omega$ if load were halved what will be the speed?

(2) $\omega = \frac{200}{(k_e\phi)} - \frac{2 \times 25/2}{(k_e\phi)} = \frac{175}{(k_e\phi)}$

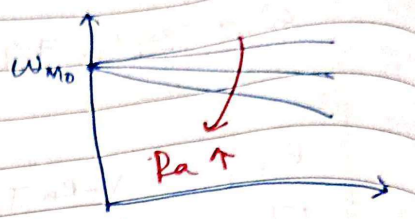
$\omega = \frac{175}{150} \times 500 = \boxed{460.53 \text{ RPM}}$

(1) $\omega = \frac{200}{(k_e\phi)} - \frac{(25/2)(0.4)}{(k_e\phi)} = \frac{195}{190} \times 500$

$\omega = \frac{500}{190} \times 187.5 = \boxed{513.16 \text{ RPM}}$

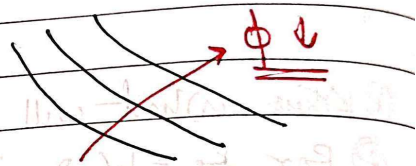


Input Voltage Control

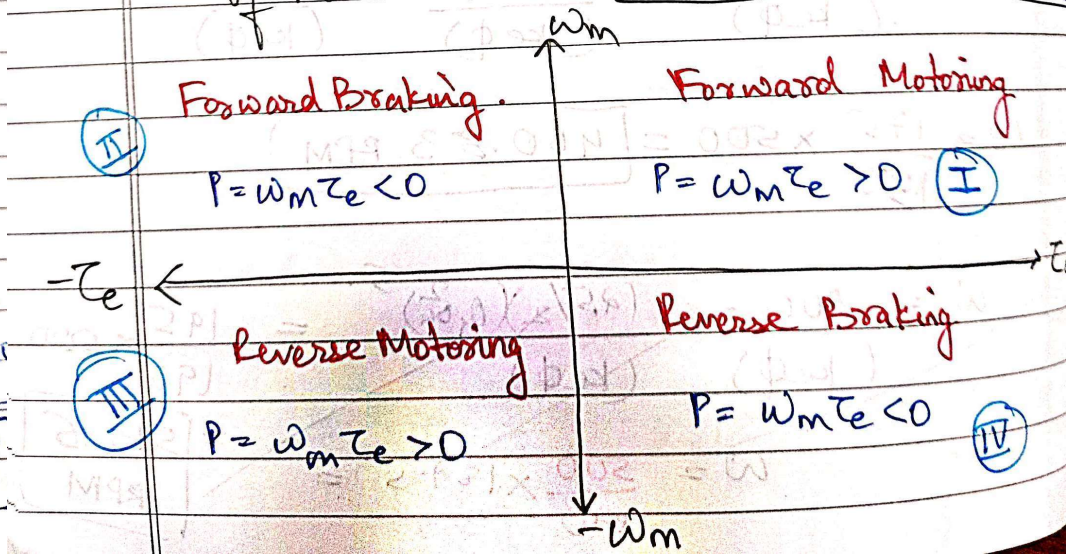


Armature resistor control

Field flux control
AKA
field control.



4 Quadrant Operation of Machine.



$\tau_e = (k_e \phi) I_a$ } Can be made -ve by either
① Reversing dir of flux (ϕ)
② Reversing dir of I_a current.

Braking ⇒ Electrical Braking.

$V = E_b + R_a I_a$.

$VI_a = E_b I_a + R_a I_a^2$

P_{in} — $E_b I_a$ — P_{arm} ⇒ Armature losses.

$E_b I_a$ ⇒ Output Power (Mechanical).
P_{mech}

P_{mech} : Gross mechanical Output power : $P_{gross,mech}$

$P_{gross,mech} = E_b I_a$.

~~Electrical~~

⇓

Electrical Equivalent of the Mechanical Power.

⇓

Without taking losses into account
∴ Gross

Monday

Self excited Motors get initial flux due to residual magnetism, more flux flux due to the feedback.

D.C. Machines → Generator, Motor

(a) SEDC Motor
$I_L = I_a$

D.C. Shunt Motor
$I_L = I_f + I_a$

D.C. Series Motor
$I_L = I_f = I_a$

D.C. Compound Motor
$I_L = I_{f1} + I_{f2}$
$I_{f2} = I_a$

(b) Self-excited motors

(a) SEDC Generator

D.C. Shunt Gen.

D.C. Series Gen.

D.C. Compound Gen.
$I_L = I_{f1}$
$I_{f2} = I_a - I_{f1}$

(b) self-excited generators.

NOTE: For compound Motor or Gen, above variation are possible.

Ques

4A @ No load
$I_f = \frac{250}{250} = 1A$
250V
$R_f = 250\Omega$
$R_a = 0.8\Omega$

→ $V_t I_{L0} = 250 \times 4 = 1000W$

→ $I_{a0}^2 \times R_a$ = No Load Arm Cu losses
$I_{a0} = I_L - I_f = 4 - 1 = 3A$; $V_f I_f = 250W$
$\therefore I_{a0}^2 R_a = 3^2 \times 0.8 = 7.2W$

→ $(V_t I_{L0} - V_L I_f) = P_{C0} = 1000 - 250 \times 1$
$= 750W$

P_{wf} → Windage and friction loss
$P_{core} + P_{wf} = P_{A0} - I_{a0}^2 R_a = 750 - 7.2 = 742.8W$

* Load in DC Motor generally correspond to Torque, Te.
* Ke changes in field regulator case.

PageWork

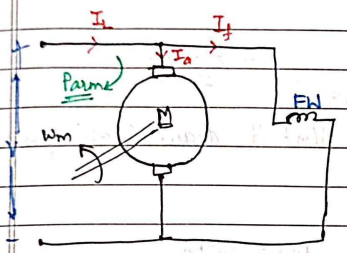
$P_{const.} = P_{fg} + P_{core} + P_{fw.} = 742.8\,W + 250\,W = 992.8\,W$

$P_{variable} = I_a^2 R_a$

$\eta_m = \frac{P_o}{P_{in}} = \frac{V_L I_L - (P_{const.} + P_{variable})}{V_L I_L} \times 100\%$

$\eta_g = \frac{P_o}{P_{in}} = \frac{V_L I_L}{V_L I_L + P_{const.} + P_{variable}} \times 100\%$

Tuesday 28/02/2023



$P_{arm} = P_{wf} + P_{iron}$ — (1) ; $P_{f,cu} = V_f I_f$ or $I_f^2 R_f$

$P_{total} = P_{fg} + P_{iron} + P_{f,cu} + I_a^2 R_a$

$= P_{const} + P_{var}$

We can find η at some load without even connecting load.

$V I_{a0} - I_{a0}^2 R_a = P_{arm}$ — (1)

$\Rightarrow P_{const} = (V I_{a0} - I_{a0}^2 R_a) + V_f I_f$ — (A)

$P_{var} = I_a^2 R_a$ — (B)

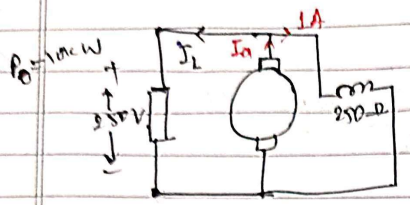
(A) & (B)

$P_{total} = (A) + (B)$

For previous e.g., $P_{const} = 742.8 + 250 = 992.8\,W$

All this power is just utilised to overcome losses (992.8 W out of 1000 W); no useful power output

Q.E: 10 kW power at 250 V, generator with specifications as given.



$I_L = \frac{10k}{250} = 40\,A$

$\therefore I_a = 41\,A$

$P_{var} = I_a^2 R_a$

$= 41^2 \times 0.8$

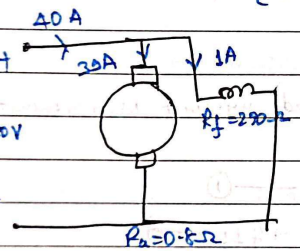
$P_{var} = 1334.8\,W$

$P_{const.} = 992.8\,W$

$P_{total} = 992.8 + 1334.8 = 2337.6\,W$

$\therefore \eta_g = \frac{10 \times 10^3}{10 \times 10^3 + 2337.6} = 81.05\%$

Q.E: If it was a motor such that it draws 10 kW power from mains, what is the η?



$P_{var} = I_a^2 R_a$

$= 39^2 \times 0.8$

$= 1216.8\,W$

$P_{const} = 992.8\,W$

$\therefore \eta = \frac{10 \times 10^3 - (992.8 + 1216.8)}{10 \times 10^3} = 77.9\%$

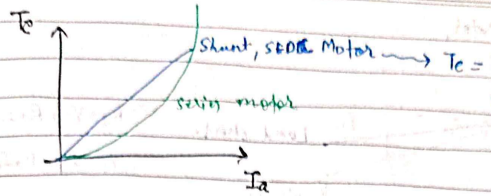
Swinburne's Test (@ No Load): Pconst calculated at no load condition can be exploited to calculate efficiency at any given load.

Motor:

$V_{in} = E_b + R_a I_a = K_e \phi \omega + R_a I_a$

$\Rightarrow \omega = \frac{V_{in} - R_a I_a}{K_e \phi} = \frac{V_{in}}{K_e \phi} - \frac{R_a}{(K_e \phi)^2} T_e$

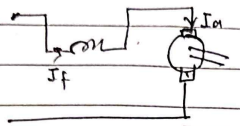
$T_e = K_t \phi I_a$



$T_e = K_e \phi I_a$

$\propto I_a$

Ref for series DC Motor

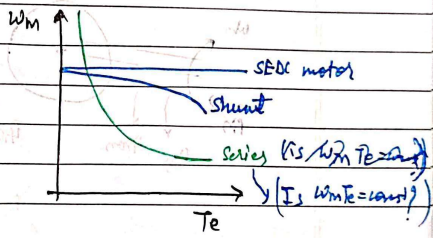


$T_e = K_e \phi I_a$

$\propto I_f I_a$

$\boxed{T_e \propto I_a^2} \quad (I_f = I_a)$

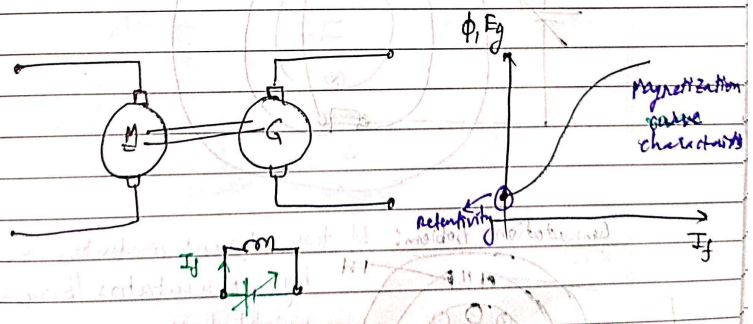
Now, $\omega_m = \frac{V}{K_e \phi} - \frac{R_a T_e}{(K_e \phi)^2}$



For series DC Motor

$\omega_m \propto \frac{V}{I_a} - C$

$\Rightarrow$ If series Motor is started without load, then speed is too high, thus it should not be started without load.

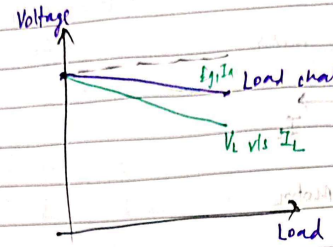


$E_g = K_e \phi \omega_m$

$\propto (K_e \omega_m) \phi$

$\propto C_1 I_f$

For generator,



$E_g = V_L + R_a I_a$

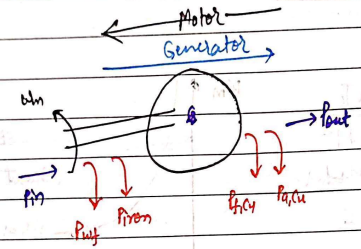
$V_t = E_g - R_a I_a$

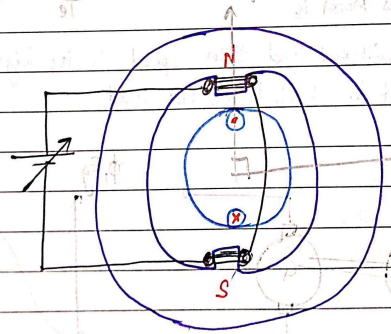
$E_g, I_a \rightarrow$ Internal char.

$V_L, I_L \rightarrow$ External char

Wednesday

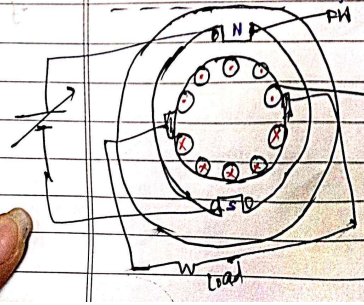
01/03/23





* $F_f \perp F_a$ irrespective of ωm

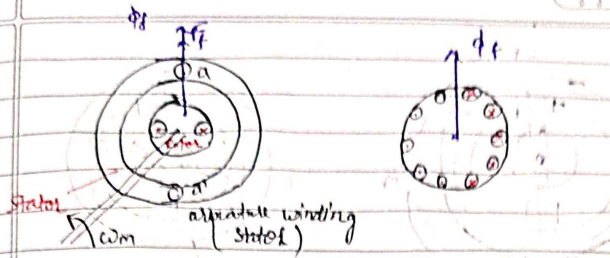
Commutation Problem: If two adjacent conductors are shorted by commutators / brushes, sparks might happen.



Arm. Power > Field Power

(∴ Bulk power)

Mechanical power inputs of Generator: Water, Steam (from coal) etc.



⇒ f_f is moving with rotor as field windings are constructed on rotor.

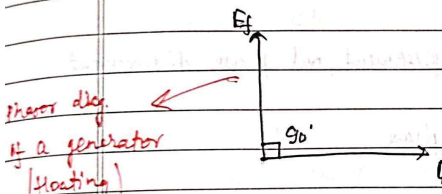
$$\phi = \phi_m \sin \omega t$$

$$e = -N \frac{d\phi}{dt}$$

$$e = -N \phi_m \omega \cos \omega t = N \phi_m \omega \sin(\omega t + \pi/2)$$

$$= 2\pi f \phi_m N \sin(\omega t + \pi/2)$$

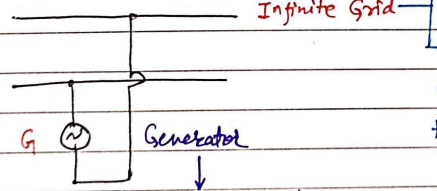
$$\Rightarrow E_{(rms)} = \sqrt{2}\pi f \phi_m N \text{ ; Volts}$$



Phasor diag. of a generator (floating)

for the case of TIF, E will lag ϕ_f.

(Load convention??)



→ Infinite source of power

→ Grid Voltage doesn't change irrespective the no. of loads attached [No Loading Effect]

floating on the grid

⇓

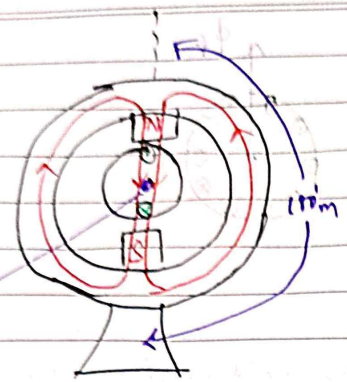
Neither doing or receiving power from grid

known as

↳ This is AC machine i.e. when armature → stator, field winding → rotor.

↳ Advantage of this is that tearings due to brush contacts will be less as field winding there has relatively lower current

Fitzy



2-pole Machine

4-pole Machine

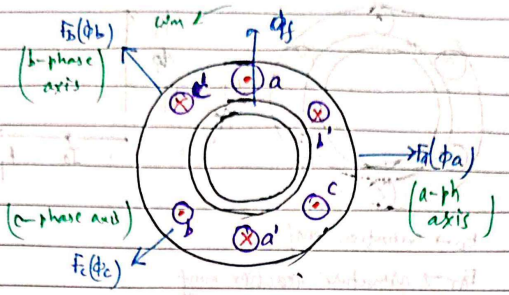
↳ Also there is 180° phase shift blw armature conductors i.e. if one has induced emf B across if the other has -B.

Mechanical angle: Space displacement, not phase displacement (θ_m)

Phase angle: Difference in phase

P → no. of poles.

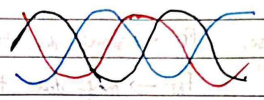
f: frequency of magnetic reversal (in Hz)



$e_a = E_m \sin(\omega t)$

$e_b = E_m \sin(\omega t - 120^\circ)$

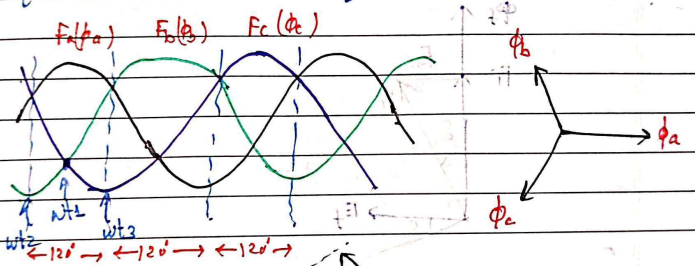
$e_c = E_m \sin(\omega t - 240^\circ)$



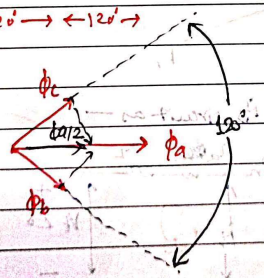
↳ ϕ_a, ϕ_b & ϕ_c are sinusoidal in nature.

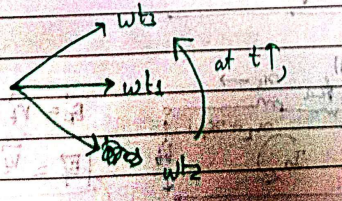
[∵ e_a, e_b & e_c are sinusoidal]

∴



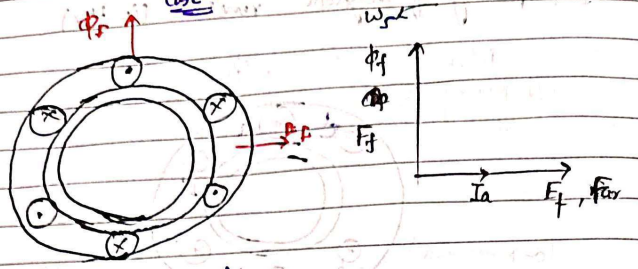
@ ωt_1





Monday

Case-1: I_a is in phase with E_f



E_f → excitation emf

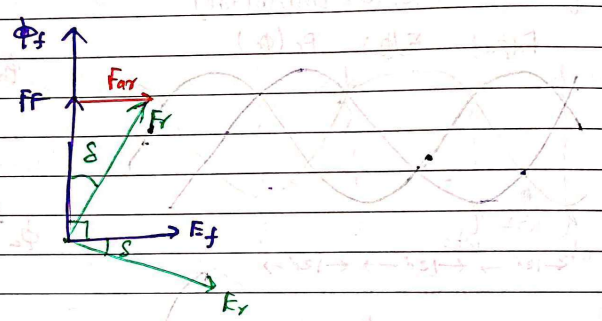
F_{ar} → armature reaction mmf

Also, F_f → mmf due to field current (Rotor)

F_{ar} → mmf due to load current (Stator)

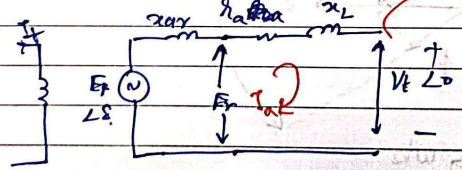
F_r → resultant mmf (air gap mmf)

δ → load angle / power angle

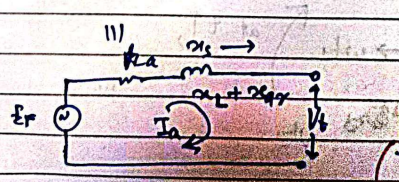


Thus we can model its circuit as —

1-ϕ eq. circuit of alternator.



3-ϕ Generator or Alternator or Synchronous Generator

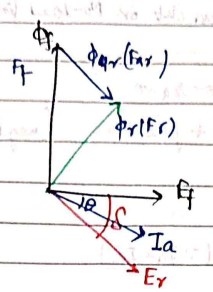


$\vec{E_f} = \vec{V_t} + \vec{I_a} Z_s$ → Synch. Impedance

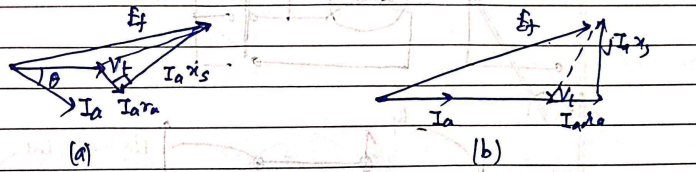
$\vec{E_f} = \vec{V_t} + \vec{I_a}(r_a + j x_s)$

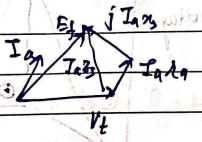
This eqn. is similar to TIF eqn.

Case-2: If I_a lags the voltage by angle θ.



∴ Similar to T/F, we can have 3-types of T/F.





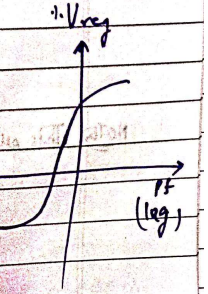
Ratings VA, kVA, mVA

$$\Rightarrow E_f = \sqrt{(V_t\cos\theta + I_a r_a)^2 + (V_t\sin\theta \pm I_a x_s)^2}$$

(+ ↓ for lag, − ↓ for leading)

Voltage regulation of the alternator:

$$\%\,Reg = \frac{E_f - V_t}{V_t} \times 100\%$$



@ No load;

Open ckt: $\overline{V_t} = \overline{E_f}$ → Open ckt or No-load Voltage

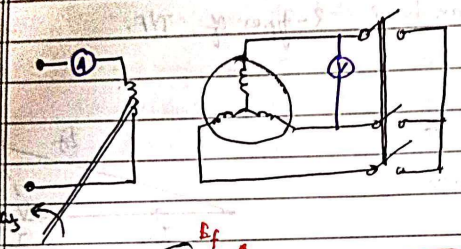
Now for synch. machines, Resistance R_a no.

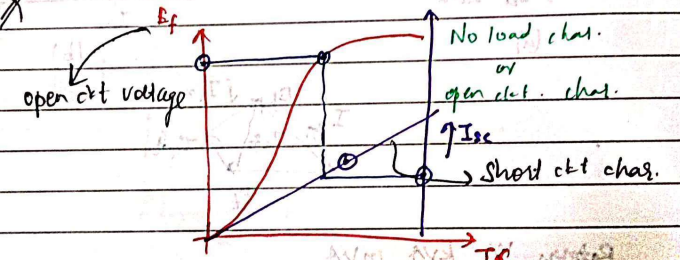
Short ckt: $\therefore \overline{E_f} = \overline{V_t} + j\overline{I_a} x_s$ [negl. R_a]

$x_s = x_{ar} + x_L$

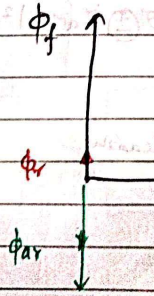
↳ for $V_t = 0$; $\Rightarrow I_a x_s = E_f$

$\Rightarrow x_s = \frac{E_f}{I_{sc}}$ [$\because I_a \equiv I_{sc}$]





"MAGNETISATION CHARACTERISTICS"



⇒ ϕ_r is small

⇒ ckt never gets into saturation

⇒ I_{sc} Vs I_f is linear

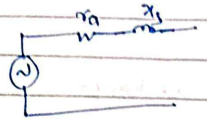
NOTE: This all is done for equivalent 1-ϕ ckt

2-φ Qtns from ECN

Monday

→ Rated output power → Volt across 1-φ

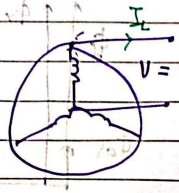
QUE: A 500 kVA, 3.3 kV, 3-φ, Y-connected alternator, $r_a = 0.3\ \Omega$ & $x_s = 4.0\ \Omega$. Determine full load current.



DC Motor → Power in Watts

Synchronous → Power in Volt-Amps

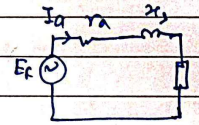
ANS: a) $\sqrt{3} V_L I_L = 500 \times 10^3$



$$I_L = \frac{500 \times 1000}{\sqrt{3} \times 3.3 \times 10^3} = 87.5\text{A} = I_{rated}$$

$$\therefore V_t = \frac{3.3}{\sqrt{3}}\text{ kV} = 1905.25\text{ V}$$

b) For Δ systems ~~at~~ At full load 0.8 pf lag, find open ckt voltage of alternator.



$$\bar{E}_f = \bar{V}_t + \bar{I}_a(r_a + jx_s)$$

lagging pf

$$E_f = \sqrt{(V\cos\phi + I_a R_a)^2 + (V\sin\phi \pm I_a x_s)^2}$$

leading pf

Using E_f for 0.8 leading pf

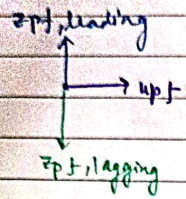
$$\therefore E_f = 500 \times 0.8 \sqrt{\left(\frac{3.3\times10^3}{\sqrt{3}} + 87.5 \times 0.3\right)^2 + \left(\frac{3.3\times10^3}{\sqrt{3}} \times 0.6 + 87.5 \times 4\right)^2}$$

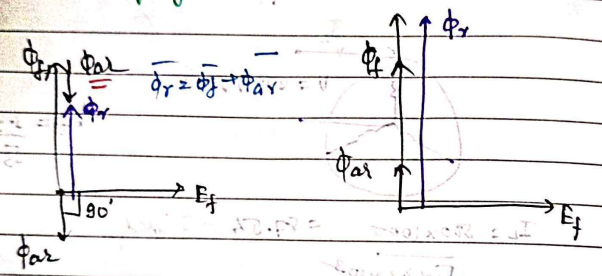
$$\Rightarrow E_f = 2152.5\text{ V}$$

So, $\%V_{reg} = \frac{E_f - V_t}{V_t} \times 100\% = \frac{2152 - 1905}{1905} = 12.96\%$

$f = \frac{(P)Nm}{120}$

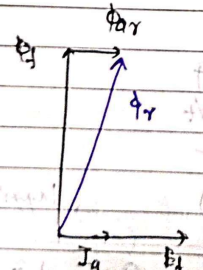
For Diff. phases of I_a, we have—





(a) zpf lagging (Demagnetization)

(b) zpf leading (Magnetizing)



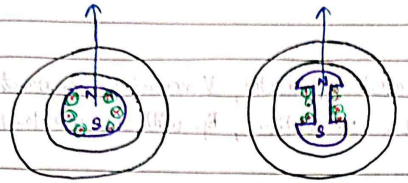
(c) Upf (Cross magnetizing)

↳ Any other pf will be partly magnetizing & cross magnetizing or partly demagnetizing & cross-magnetizing

V_f → Excitation Voltage

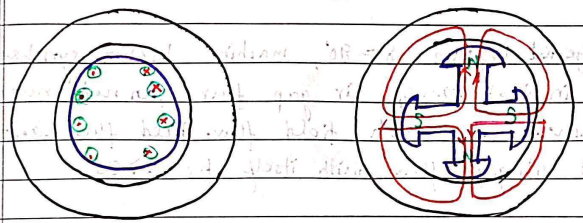
Tuesday

PageWork 14/03/2023



Cylindrical rotor	Salient Pole rotor
↳ Long axle	↳ Moderate axle
↳ Small diameter	↳ Large diameter i.e. large rotor
↳ Lakhs per rpm	↳ Used in power generating station
↳ Low to medium power	↳ Rated for high power, low speed applications
	↳ Contains projected poles
↳ Uniform air gap	↳ Non uniform air gap

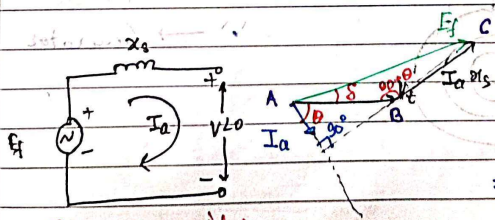
2-pole Structures



4-pole structures

Q Why more multiple pole systems are used?

⇒ Use $f = \frac{pN_m}{120}$ where N_m is in rpm, we can regulate for N_m by varying p as per our requirement



For ΔABC,

$$\frac{E_f}{\sin(90+\theta)} = \frac{I_a x_s}{\sin\delta}$$

$$\Rightarrow \frac{E_f \sin\delta}{x_s} = I_a \cos\theta$$

Ignore r_a as it is small most of alternators

$$\Rightarrow \underbrace{\frac{3VE_f \sin\delta}{x_s}}_{P_{mech} \text{ or } P_{in}} = \underbrace{3VI_a \cos\theta}_{\text{Electrical power o/p } (P_e \text{ or } P_{out})}$$

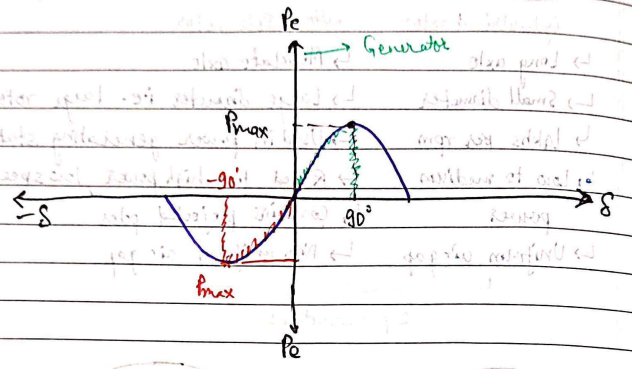
Load ↑, I_a ↑, δ ↑ ← δ is equivalent to load

NOTE: We can run a prime mover without a load connected to it.

PageWork

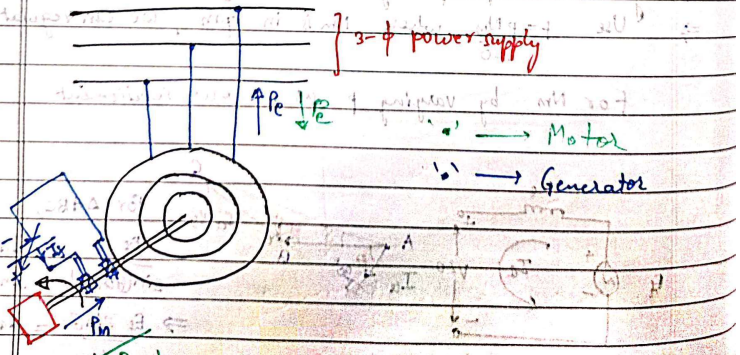
Now, $P_e = \frac{3VE_f \sin\delta}{x_s}$

For a given rated alternator, V remains constant And for a given excitation, E_f will be constant too. So, we have -



↳ Beyond δ = 90°, machine loses its synchronism as in that range, air gap flux can not move synchronously with field flux. Field flux can't pull air gap flux with itself for δ > 90°

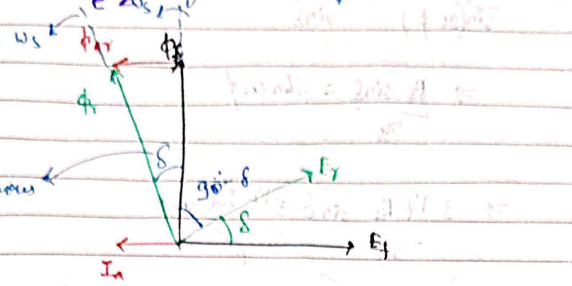
Floating alternator



Stray losses ——?——> can't be determined.

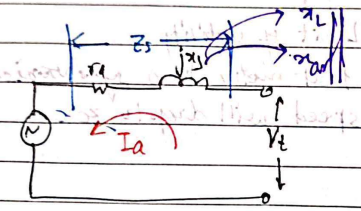
Synch. Motor

If direction of I_a is changed,



↳ Here, instead of ϕ_f driving ϕ_r; as ϕ_r is leading ϕ_f, ϕ_r will drive or pull ϕ_f.

Wednesday 15/03/23



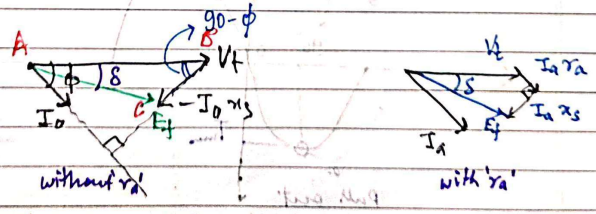
Direction opp. to synch. generator

$$\bar{V}_t = \bar{E}_f + \bar{I}_a Z_s$$

$$\bar{V}_t = \bar{E}_f + \bar{I}_a (r_a + jx_s)$$

$$= \bar{E}_f + j\bar{I}_a x_s$$

[$\because r_a$ is small]



Thus • E_f is less than E_f of Generator
• E_f is lagging here, but it was leading before [V_t is fixed]

These conclusions are valid even if r_a is considered.

For ΔABC, $\frac{E_f}{\sin(90+\phi)} = \frac{I_a x_s}{\sin\delta}$

$$\Rightarrow \frac{E_f \sin\delta}{x_s} = I_a \cos\phi$$

$$\Rightarrow \underbrace{\frac{3 V_t E_f \sin\delta}{x_s}}_{P_{mech}} = \underbrace{3 V_t I_a \cos\phi}_{P_{in}}$$

↳ For $\delta \geq 90^\circ$, ϕ_r won't be able to drag ϕ_f synchronously due to which motor collapse.

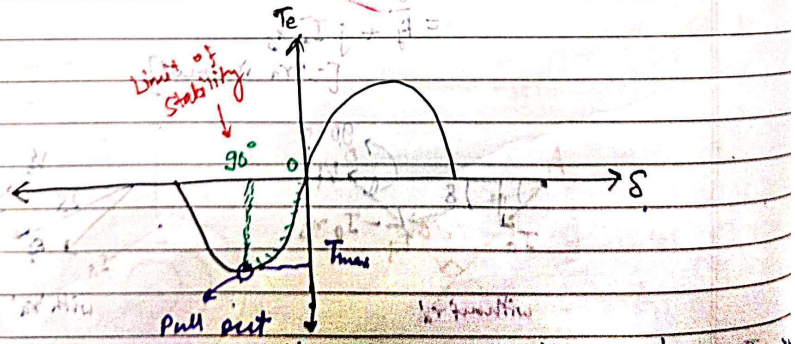
$\therefore \delta = 90^\circ$ ⟶ Limit of stability

↳ Beyond that motor loses synchronism & speed will drop to zero.

Friday 17/03/23

$$P_m = \frac{3VE_f \sin\delta}{x_s}$$

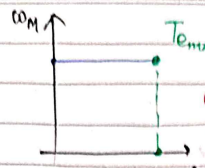
$$\therefore T_e = \frac{P_m}{\omega_s} = \frac{3VE_f \sin\delta}{\omega_s x_s}$$

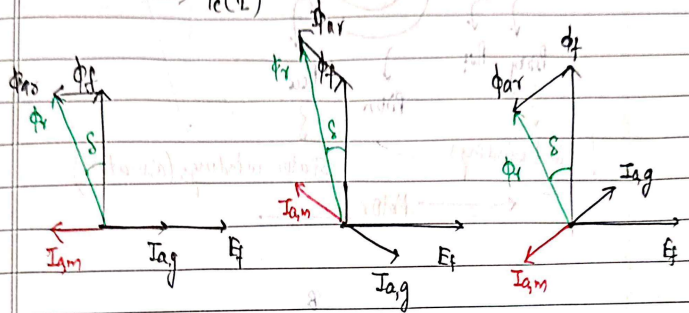


Torque beyond this machine 'pulls out' of synchronism

$N_s = \frac{f}{P} \times 120$

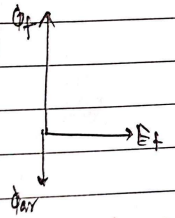
Synchronous speed

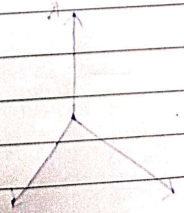


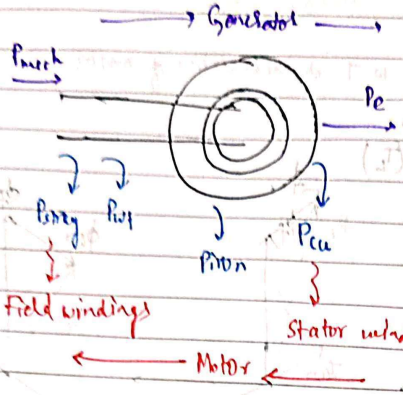


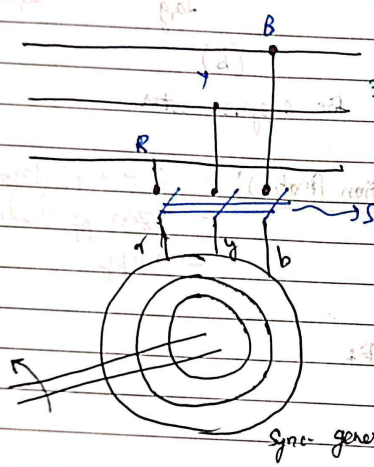
For a sync. motor

Armature Reaction (Motor): → zero pf (lagging)
→ zero pf (leading)
→ upf

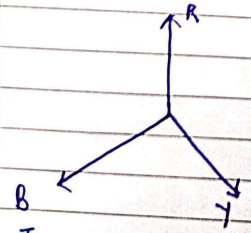


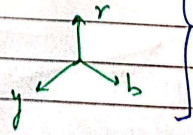






↳ What happens when connection is made with wrong phases?





⇒ Sparkling will happen

∴ To avoid this, regulate excitation excitation current such that phase voltages are same for both ckts.

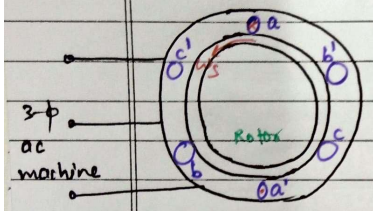
Induction Machines

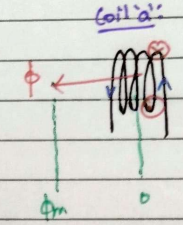
[Asynchronous Machines]

Induction Motors

↳ Asynch. Motors are quite popular as compared to Asynch. Gen.

↳ Speed of the mag-field & speed of the rotor not same, that's why its called asynchronous.





↳ ϕ will be pulsating with time

↳ ϕ will also vary sinusoidally along its cross section.

Initially → stationary rotor

W_s: freq. of rotation of mag field.

In the direction of coil which carries max current.

$$W_s = \frac{4\pi f}{P}$$ f → freq of a.c.

$$N_s = \frac{120 f}{P}$$

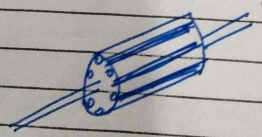
Types of rotor:

① Squirrel Cage

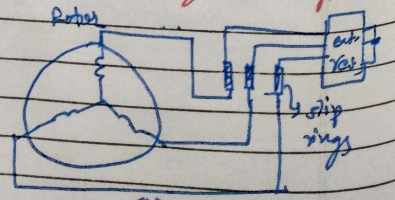
② Slip-ring (wound rotor)

Can't add resistance to rotor

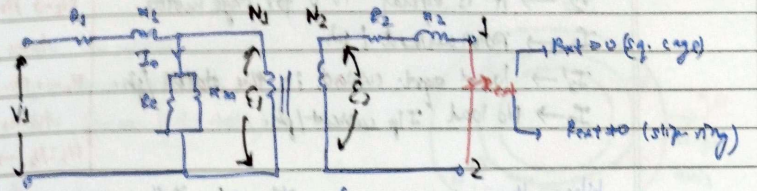
Resistance can be connected through slip rings



Squirrel Cage motor



Slip Ring Motor



↳ Above similarly for IM & TP is valid if rotor is stationary [i.e. Standstill condition]

↳ No resistance across 12.

* HOW DOES INDUCTION MOTOR STARTS?

⇒ Due to em induction, rotor develops current s.t. it opposes ϕ due to stator, this current develops an mmf which gets repelled or attracted by ϕ already exists due to stator. Due to this repulsion or attraction, rotor starts rotating.

$$\text{Slip} = s = \frac{W_s - W_m}{W_s}$$

SW_s → slip speed $= W_s - W_m$

$$\Rightarrow \boxed{W_m = W_s(1-s) \text{ ; rad/s}}$$

$$\Rightarrow \boxed{N_m = N_s(1-s) \text{ ; RPM}}$$

↳ $W_s - W_m$ should be non zero, because if it were zero, induced emf will be zero due to which rotor won't rotate.

V_1 → Stator input voltage/phase

E_1 → Stator induced emf/ph

R_1 → Stator winding res./ph

X_1 → Stator leakage reac/ph.

R_c/X_m → mag. imp./ph

R_2 → Rotor winding res./ph

X_2 → Rotor leakage reac./ph

$\frac{E_2}{E_2'} = \frac{\omega_s}{\omega_s - \omega_m} = \frac{1}{s}$

$s=1, \quad E_2 = E_2$

$s=s, \quad E_2' < E_2$

$E_2 \propto \omega_s - \omega_m$

$V_2 \rightarrow$ It is equal to '0' for cage motor

$I_2 \rightarrow$ Rotor current / ph.

$I_2' \rightarrow$ load cqvt. current in the stator / ph.

$I_0 \rightarrow$ No load I/p current / ph.

$N_1 \rightarrow$ No. of stator turns / ph.

$N_2 \rightarrow$ No. of rotor turns / ph.

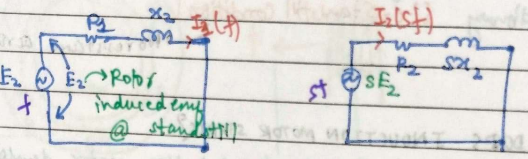
$N_1 : N_2 \rightarrow$ turns ratio

When the rotor runs with slip 's'.

Rotor speed is ω_m,

$s\omega_s = \omega_s - \omega_m$

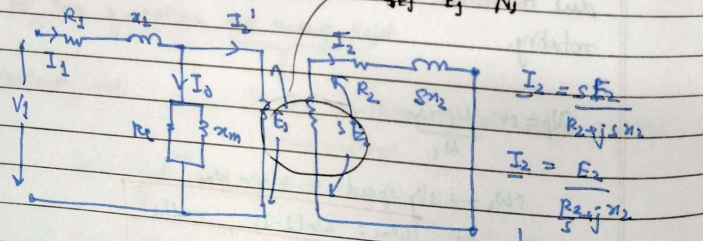
$\omega_m = \omega_s(1-s)$



@ standstill (rotor stationary)

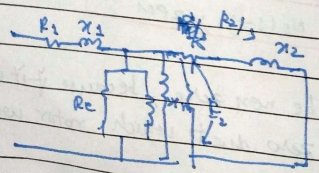
@ any speed ω_m or 's'

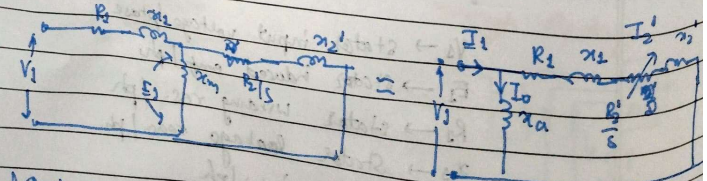
$\therefore \frac{sf_1}{sf_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1}$



$I_2 = \frac{sE_2}{R_2 + jsx_2}$

$I_2 = \frac{E_2}{\frac{R_2}{s} + jx_2}$





Simplified eq. ckt of the 3-ϕ IM (Approx. eq. ckt)

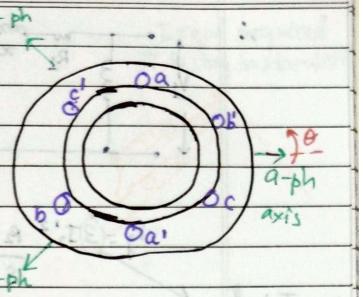
↳ Here, I_0' will be much more than I_0 of T/F.

$\omega_m = \frac{4\pi f'}{P} \quad \& \quad \omega_s = \frac{4\pi f}{P}$

$i_a = I_m \cos(\omega t)$

$i_b = I_m \cos(\omega t - 2\pi/3)$

$i_c = I_m \cos(\omega t - 4\pi/3)$



$\therefore F_a = i_a N_{ph} \cos\theta$

$= I_m N_{ph} \cdot \cos\omega t \cos\theta$

$F_b = I_m N_{ph} \cos(\omega t - 2\pi/3) \cos(\theta - 2\pi/3)$

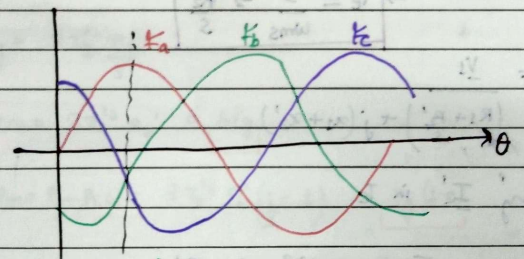
$F_c = I_m N_{ph} \cos(\omega t - 4\pi/3) \cos(\theta - 4\pi/3)$

$\therefore F_{res} = F_a + F_b + F_c$

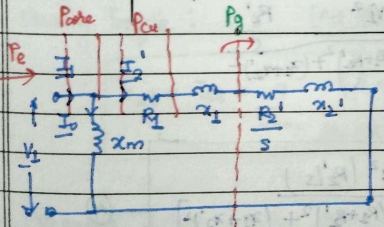
$= \frac{I_m N_{ph}}{2} [\cos(\omega t + \theta) + \cos(\omega t - \theta) + \cos(\omega t + \theta - 4\pi/3) + \cos(\omega t + \theta - 2\pi/3)]$

$F_{res} = \frac{3}{2} I_m N_{ph} \cos(\omega t - \theta)$

displacement in phase — space displacement



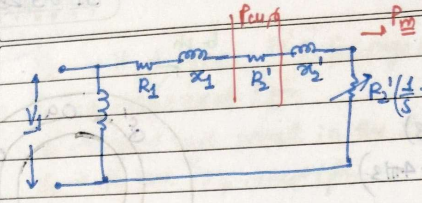
Now, P_e



$I_1 = I_0 + I_2'$

Simplified eq. ckt of 3-ϕ IM (appro. eq. ckt)

$\frac{R_2'}{s} \Leftrightarrow R_2' + R_2'\left(\frac{1}{s} - 1\right)$



$\therefore$

$P_g = (3)I_2'^2 \frac{R_2'}{s}$

For Total 3 phases $P_{r,cu} = 3I_2'^2 R_2'$

$\Rightarrow P_g - P_{r,cu} = P_m = 3I_2'^2 \frac{R_2'}{s}(1-s) = 3I_2'^2 R_2'\left(\frac{1}{s}-1\right)$

$\therefore T_e = \frac{P_m}{\omega_m} = \frac{3 I_2'^2 \frac{R_2'}{s}(1-s)}{\omega_{ms}(1-s)}$

$\Rightarrow \boxed{T_e = \frac{3}{\omega_{ms}} I_2'^2 \frac{R_2'}{s}}$

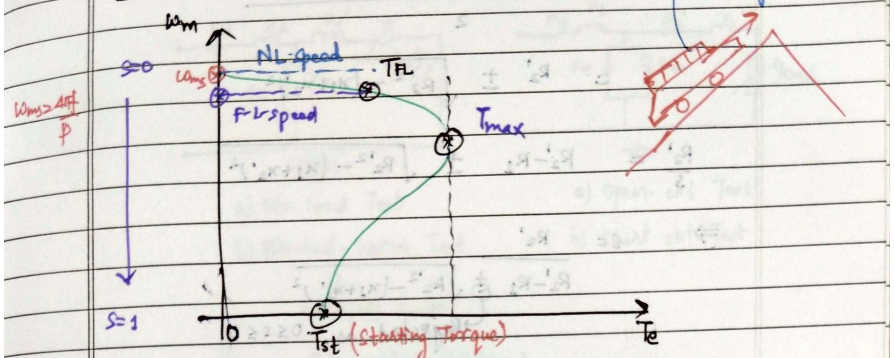
Also, $I_2' = \frac{V_1}{\left(\frac{R_1+R_2'}{s}\right) + j(x_1+x_2')}$

Putting I_2' in T_e

$$T_e = \frac{3}{\omega_{ms}} \frac{V_1^2 \frac{R_2'}{s}}{\left(\frac{R_1+R_2'}{s}\right)^2 + (x_1+x_2')^2}$$

$$\boxed{T_e = \frac{3}{\omega_{ms}} \frac{V_1^2 (R_2'/s)}{\left[\left(\frac{R_1+R_2'}{s}\right)^2 + (x_1+x_2')^2\right]}} \quad \text{(A)}$$

Torque - speed Characteristics:



$T_e = T_L + J\frac{d\omega_m}{dt} \Rightarrow T_e - T_L = J\frac{d\omega_m}{dt}$

$\therefore$ Change of the speed from No load to Full load is very less, IM is known as constant speed motor.

Now, $P_g = 3I_2'^2 \frac{R_2'}{s}$

$P_{r,cu} = 3I_2'^2 R_2' = sP_g$

$\therefore P_m = P_g - P_{r,cu} = 3I_2'^2 R_2'\left(\frac{1}{s}-1\right) = P_g(1-s)$

↳ Starting Torque : $s=1$ [Put this in (A)]

$$\therefore \boxed{T_{e,st} = T_{st} = \frac{3}{\omega_{ms}} \frac{V_1^2 R_2'}{(R_1+R_2')^2 + (x_1+x_2')^2}}$$

↳ Maximum Torque / Break Torque

$$\frac{dT_e}{ds} = 0 \Rightarrow \frac{1}{s}\left[2\left(\frac{R_1+R_2'}{s}\right)\cdot R_2'\cdot\left(+\frac{1}{s^2}\right)\right] = \left[\left(\frac{R_1+R_2'}{s}\right)^2 + (x_1+x_2')^2\right]\left(\frac{+1}{s^2}\right)$$

$$\Rightarrow 2R_2'\left(\frac{R_1+R_2'}{s}\right) = (x_1+x_2')^2 + \left(\frac{R_1+R_2'}{s}\right)^2$$

$$\Rightarrow \left(\frac{R_1+R_2'}{s}\right)^2 - 2R_2'\left(\frac{R_1+R_2'}{s}\right) + (x_1+x_2')^2 = 0$$

$$\Rightarrow \frac{R_1 + R_2'}{s} = \frac{2R_2' \pm \sqrt{4R_2'^2 - 4[(x_1+x_2')^2]}}{2}$$

$$= R_2' \pm \sqrt{R_2'^2 - (x_1+x_2')^2}$$

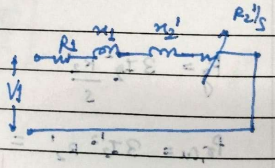
$$\frac{R_2'}{s} = R_2' - R_1 \pm \sqrt{R_2'^2 - (x_1+x_2')^2}$$

$$\Rightarrow s = \frac{R_2'}{R_2' - R_1 \pm \sqrt{R_2'^2 - (x_1+x_2')^2}}$$

↳ Rejected as $0 \le s \le 1$

$$\therefore s = \frac{R_2'}{R_2' - R_1 + \sqrt{R_2'^2 - (x_1+x_2')^2}}$$

OR Apply max. power transfer theorem:-



$$\frac{R_2'}{s_m} = |R_1 + j(x_1+x_2')|$$

$$\Rightarrow s_m = \pm \frac{R_2'}{\sqrt{R_1^2 + (x_1+x_2')^2}}$$

+ → Motoring operation

− → Braking operation

↳ Putting s_m in 'A', we get −

$$T_{max} = \frac{3}{\omega_{ms}} \frac{V_1^2 \cdot (R_2'/s_m)}{\left(R_1 + \frac{R_2'}{s_m}\right)^2 + (x_1+x_2')^2}$$

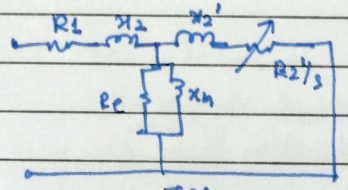
$$= \pm \frac{3}{\omega_{ms}} \frac{V_1^2}{\pm\sqrt{R_1^2 + (x_1+x_2')^2} + \sqrt{R_1^2 + (x_1+x_2')^2}}$$

$$T_{max} = \frac{3}{2\omega_{ms}} \frac{V_1^2}{R_1 + \sqrt{R_1^2 + (x_1+x_2')^2}}$$

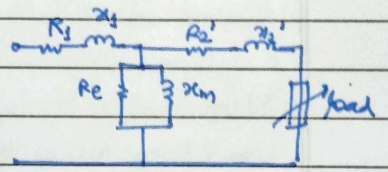
T_{max} doesn't depend on rotor windings resistance, however s_m depends on it does.

⇒ By adding R_2', we can get same T_{max} for lesser value of I_2'.

Equivalent circuit parameters of I.M. :



I.M.



T/F

a) No load Test

b) Blocked-rotor Test

↓

Standstill condn.

a) Open-ckt Test

b) Short ckt Test

$$\frac{R_2'}{s} = R_2' + \underbrace{R_2'\left(\frac{1-s}{s}\right)}_{\text{Actual variable resistance}}$$

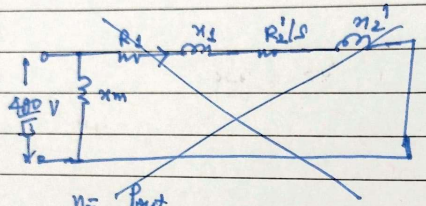
∴ For short ckt, $s = 1$.

Monday

Ques: 480 V, 50Hz, 37.3 kW, Δ connected squirrel cage IM draws I = 60A @ 0.85 pf lagging while supplying a typical load. Stator Cu losses = 2kW, rotor Cu losses = 700 W; friction & windage loss = 600W. Core / Iron loss = 1800W. Stray losses are negligible / is unknown. Find power efficiency (η).

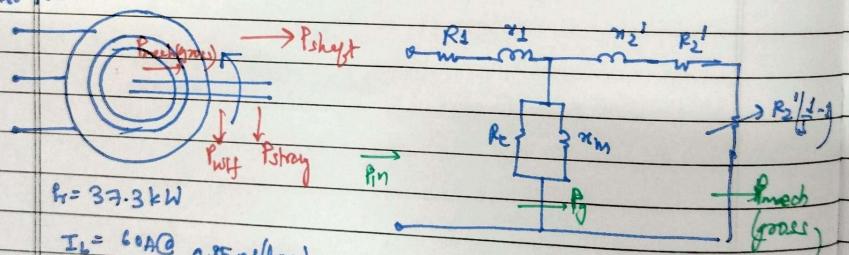
① Air gap
② Gross Mechanical pow.
③ Shaft / output power
④ η

Ans:



$\eta = \frac{P_{out}}{P_{in}}$

480 V, 50Hz



$P_T = 37.3\,kW$

$I_L = 60A\ @\ 0.85\ pf\ (lag)$

$P_{in} = \sqrt{3} V_L I_L \cos\phi = \sqrt{3} \times 480 \times 60 \times 0.85 = 42400.6\,W$

a) $P_g = P_{in} - P_{st,cu} - P_{core} = 38600.6\,W$

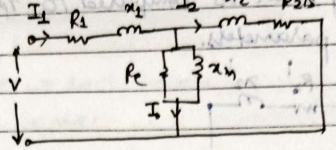
b) $P_{mech\ (gross)} = P_g - P_{cu,r} = 37900.6\,W$

c) P_{shaft} or $P_{out} = P_{mech\ (gross)} - P_{wf} - P_{stray} = 37300.6$

d) $\eta = \frac{P_{out}}{P_{in}} \times 100\% = \frac{37300.6}{42400.6} \times 100\% = 87.97\%$

More Load ↑ ⇒ slip ↑

Ques: Det. the machine parameters



Perphase eq. ckt of a 3-ϕ IM

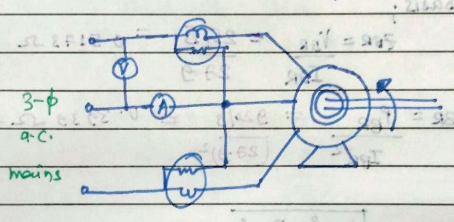
Is it because losses are small @ NL?

a) No-load Test → Rated Voltage, Rated freq. → $S_{NL} \approx 0$

b) Blocked-rotor Test → $S_{BR} = 1$ → Apply small volt. such that rated current flows.

c) Stator-resistance test

d) Rotational or Mech. loss: P_{core}, P_{wf}



7.5 HP, 4-pole
208 V, 60Hz
λ-connected IM
$I_{rated} = 28A$

a) NL Test:

Power drawn

NL ⇒ I ≈ 0.

$P_{NL} = P_{st,cu} + P_{core} + P_{wf}$ [$P_{r,cu} = 0$]

P_{rot} (Inseparable losses)

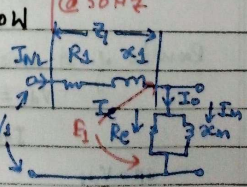
$\Rightarrow \boxed{P_{rot.} = P_{NL} - P_{st,cu}}$

Results of NL Test: $V_{NL} = 208\,V$

$I_{NL} = 8.17\,A$

$P_{NL} = 420\,W$

@ 50Hz



$E_1 = V_1 - I_1 (R_1 + j x_1)$

$\because I_{NL} = 8.17A$

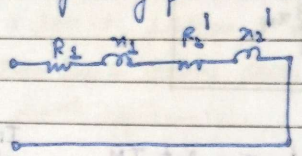
$\Rightarrow \cos\phi_0 = \frac{P_{NL}}{\sqrt{3} V_L I_L} = 0.1427 \Rightarrow \phi_0 =$

$\Rightarrow E_1 = \frac{208}{\sqrt{3}} - I_1 (R_1 + j x_1) = 118.38 \angle 0.6565°$

[R_1 & x_1 are obtained from blocked rotor Test]

b) BR Test:

⇒ Applied V_{BR} is very small as compared to rated value.
Neglect magnetizing parameters.



Results of BR Test —

$V_{BR} = 25\,V$
$I_{BR} = 27.9\,A$
$P_{BR} = 920\,W$ @ 15Hz

(25% of Rated (standard) — we have to do this to compensate skin effect.)

On per phase basis;

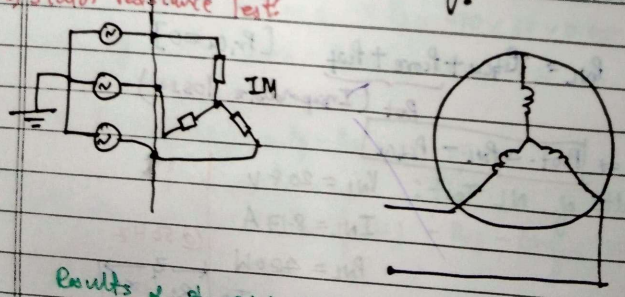
$$Z_{BR} = \frac{V_{BR}}{I_{BR}} = \frac{25/\sqrt{3}}{27.9} = 0.5173\,\Omega$$

$$R_{BR} = \frac{P_{BR}}{I_{BR}^2} = \frac{920/3}{(27.9)^2} = 0.3939\,\Omega = R_1 + R_2'$$

$$\therefore X_{BR} = \sqrt{Z_{BR}^2 - R_{BR}^2} = 0.3353\,\Omega$$

- How to determine R_1 & R_2' separately?

c) Stator resistance Test



Results of St. resist. Test —

$V_{dc} = 1.6\,V$
$I_{dc} = 2\,A$

$$2R = \frac{V_{dc}}{I_{dc}}$$

$$\Rightarrow R = \frac{V_{dc}}{2I_{dc}} = 0.243\,\Omega = R_1$$

- Most of the time IM used is class A type.
- Larger the slip, lower the Pmech ⇒ η ↓

$$\Rightarrow R_2' = R_{BR} - R_1 = 0.1509\,\Omega$$

- Now, how we can know x_1 & x_2' separately?

⇒ Since we don't know the magnetic flux leakage phenomena, we assume

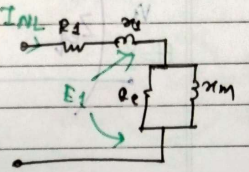
$$x_1 = x_2' = \frac{X_{BR}}{2} = 0.1677 \quad @15Hz$$

$$\Rightarrow x_1 = x_2' = 0.1677 \times \frac{60}{15} = 0.6708\,\Omega \quad @60Hz$$

↳ As calculated before, (visiting NL Test again)

$V_{ph} = 208/\sqrt{3} = 120.089\,V$

$E_1 = 118.38\,\angle 0.8565°\,V$



$\because P_{rot} = P_{NL} - P_{s.cu}$

$= 420 - 3I_{NL}^2 R_1$ (48.7)

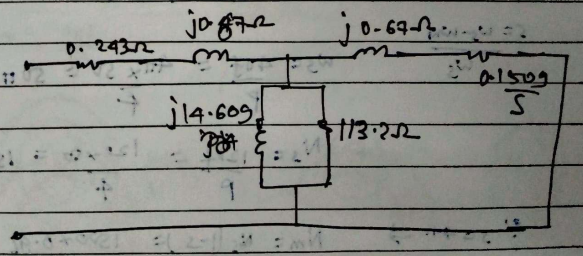
$\Rightarrow P_{rot} = 371.3\,W$

$$\therefore R_e = \frac{E_1^2}{P_{rot}} = \frac{118.38^2}{371.3}$$

$\Rightarrow R_e = 113.2\,\Omega$

$$\Rightarrow I_e = \frac{E_1}{R_e} = 1.0457\,A$$

And, $I_\mu = \sqrt{I_{NL}^2 + I_e^2} = 8.23665\,A$



PageWork 12/04/2023

Wednesday

QUE: 3ϕ 400V, Δ connected 4 pole IM has machine parameters as given —

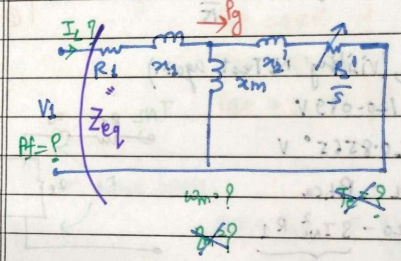
$R_1 = 0.15\Omega$, $x_1 = 0.45\Omega$

$R_2' = 0.13\Omega$, $x_2' = 0.45\Omega$ & $x_m = 25\Omega$

$s = 4\%$. Find

I_1 (line current), pf, created or Nrated, ~~T_e~~ & ~~η~~

ANS:



Since R_c not given, neglect it.

$$Z_{eq} = \frac{(s x_2' + R_2')(s x_m)}{(R_2' + s(x_2' + x_m))}$$

$$Z_{eq} = (R_1 + jx_1) + \frac{(jx_m)\left(jx_2' + \frac{R_2'}{s}\right)}{\frac{R_2'}{s} + j(x_2' + x_m)}$$

$$= 3.02 + j1.55\ \Omega$$

$$= 3.25\ \angle 21.47^\circ\ \Omega$$

$$s = \frac{\omega_s - \omega_m}{\omega_s}$$

$$\omega_s = \frac{4\pi f}{P} = \frac{4\pi \times 50}{4} = 50\pi \text{ rad/s}$$

$$N_s = \frac{120 f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$\because s = 4\% \Rightarrow$ $N_m = N_s(1 - s) = 1500 \times 0.96$

$\Rightarrow \boxed{N_m = 1440 \text{ rpm}}$

$\Rightarrow \omega_m = 50\pi(0.96) = \boxed{48\pi \text{ rad/s}}$

Shivam

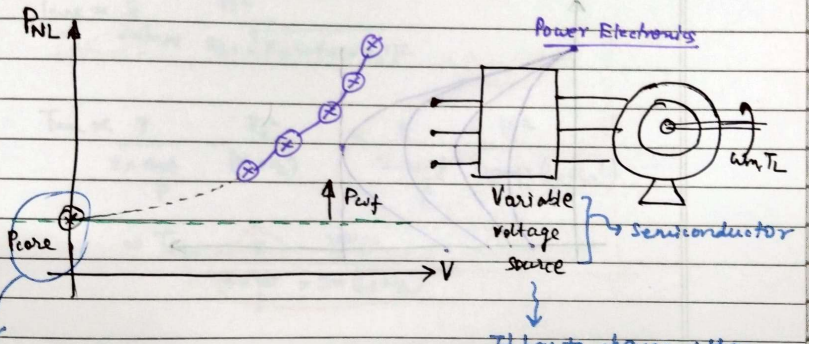
PageWork

$$I_1 = \frac{\frac{400}{\sqrt{3}}\angle 0^\circ}{3.25\ \angle 21.47^\circ} = 71.06\ \angle -21.47^\circ\ A$$

$\therefore pf = \cos(21.47^\circ)$ (lagging)

$\Rightarrow \boxed{pf = 0.93 \text{ lag.}}$ $\boxed{I_1 = 71.06\ A}$

$P_{in} = \sqrt{3} V_L I_L \cos\phi = \boxed{45,784 \text{ kW}}$

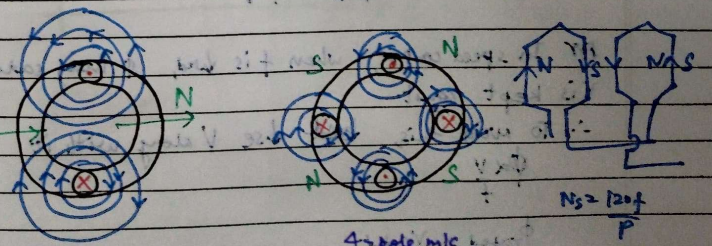


This will help us to separate determine P_{core} and thus separate P_{core} & P_{wf} out of P_{rot}.

It has to change voltage supply according to change in load.

* Speed Control of I.M.

$$\omega_{ms} = \frac{4\pi f}{P}$$



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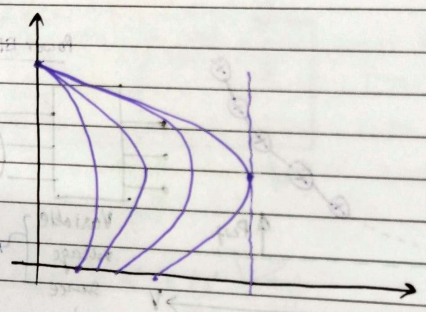
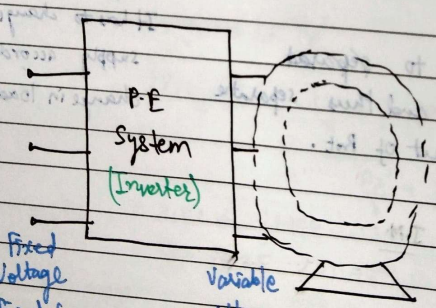
Tuesday

$$T_{max} = \frac{3}{2\omega_{ms}} \cdot \frac{V_s^2}{R_s + \sqrt{R_s^2 + (x_s + x_r')^2}}$$





$\phi \propto \frac{V}{f}$

$\omega_{ms} = \frac{4\pi f}{p}$

$\omega_m = \omega_{ms}(1-s)$

In speed control, when f is fixed, ϕ may saturate if V is kept const.

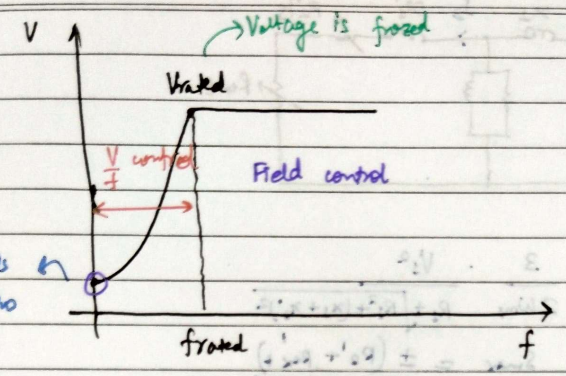
∴ To avoid this, we use V along with it.

$\phi \propto \frac{V}{f}$

$\phi_{rated} \propto \frac{V_{rated}}{f_{rated}}$

Arm. Volt-control: Below rated speed

Field Volt control: Above

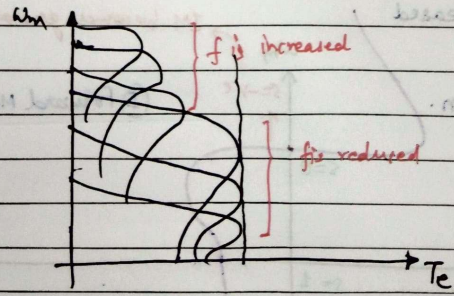


$$T_{max} = \frac{3}{2\omega_{ms}} \cdot \frac{V_s^2}{R_s + \sqrt{R_s^2 + (x_s + x_r')^2}}$$

$$T_{max} \propto \frac{3}{2 \times \frac{4\pi f}{p}} \cdot \frac{V_s^2}{(x_s + x_r')} \approx \frac{3}{2 \times \frac{4\pi f}{p}} \cdot \frac{V_s^2}{(2\pi f)(L_s + L_r')}$$

$$\Rightarrow T_{max} \approx \frac{3}{2 \times \frac{4\pi}{p} \times 2\pi (L_s + L_r')} \cdot \frac{V_s^2}{f^2}$$

$$\Rightarrow T_{max} \approx K \left(\frac{V_s}{f}\right)^2 \text{ or } K \cdot \phi^2 \quad \left[\text{Actually } \phi = \frac{E}{f}, \text{ but since } V \approx E, \phi = \frac{V}{f}\right]$$



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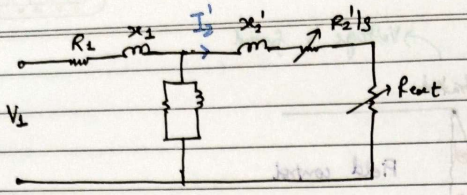
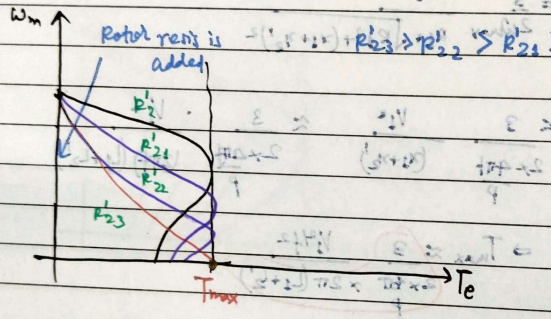


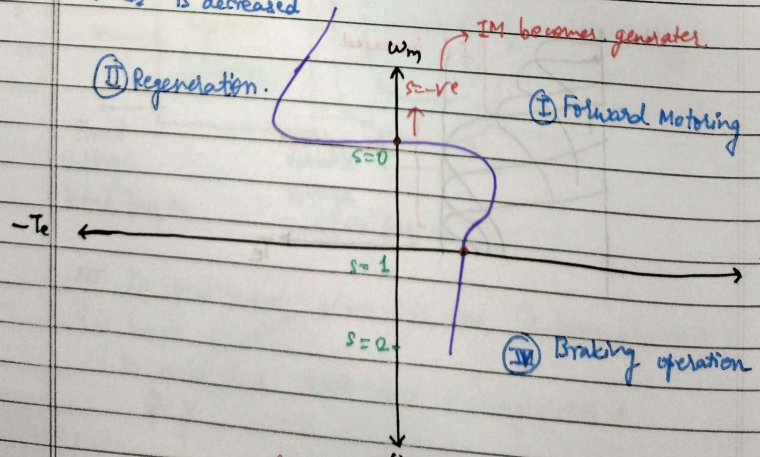
$$T_{max} = \frac{3}{2\omega_{ms}} \cdot \frac{V_1^2}{R_1 \pm \sqrt{R_1^2 + (x_1 + x_2')^2}}$$

$$S_{max} = \frac{\pm (R_2' + R_{ext}')}{\sqrt{R_1^2 + (x_1 + x_2')^2}}$$



$R_{23}' > R_{22}' > R_{21}' > R_2'$

↳ With R_{ext} added,
⇒ Sm can be controlled
→ I_2' is decreased



Complete ω-T characteristics of I.M.

If RBY is used instead of RYB, braking happens